

Introducción a Machine Learning

Autores:

**Miguel David Dávila Palomino
Diego Santiago Yáñez Chile
María Fernanda Zárate Prado**



LABORATORIO DE
**NEUROCIENCIA COGNITIVA
COMPUTACIONAL**



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Definición

Se dice que un programa de computadora aprende de la experiencia E con respecto a alguna clase de tareas T y medida de rendimiento P , si su rendimiento en las tareas en T , según lo medido por P , mejora con la experiencia E (Mitchell, 1997)



Machine Learning Algorithms

Supervised Learning

Unsupervised Learning

¿Se cuenta o no con la respuesta
correcta?

¿Cómo funciona el aprendizaje?

Regresión Lineal

Función de Coste

$$f_{(w,b)}(x^{(i)}) = wx^{(i)} + b \quad J(w, b) = \frac{1}{2m} \sum_{i=1}^m (f_{(w,b)}(x^{(i)}) - y^{(i)})^2$$

Descenso de Gradiente

$$w := w - \alpha \frac{\partial}{\partial w} J(w, b) \quad b := b - \alpha \frac{\partial}{\partial b} J(w, b)$$

¡Gracias!

ScatterPlot

- input (feature) and output (target)
- Data table is a training set
- superscripts i as a way to index a training sample
- ARTIFACT: add points to the scatter so that they can see how the scatterplot looks like.

How does the learning work?

- $f_w, b(x) = wx + b \rightarrow w$ and b are parameters (variables that can be adjust during training)
- linear regression fit a straight line to the data
- the input feature is put into the linear mode (learning algorithm) which predicts / estimates a $\hat{y}$. (y is the real true value)
- ARTIFACT: to put w and b freely and see the line

Cost function

- ARTIFACT: see bars from the line to the target y .
- equation of cost function
- cost function: subtraction and a square of the error (to avoid negative values) for all elements of the training set. To avoid getting a bigger number as the dataset increases, we calculate the mean. Debido a las reglas del cálculo, dividimos entre $2m$. Squared error cost function.
- Objective: find the w and values that minimize $J(w, b)$.
- ARTIFACT: different J values when w changes ($b = 0$). They must also see a scatter plot of J and W so that they can see the shape (concave??). Maybe with the option to just see it with two dots, and then they can also add more dots to the plot.
- ARTIFACT A: when w and b can freely vary, there is a 3D visualization of a bowl (a), a scatter plot (b) a contour plot (c), and a learning curve (d). The arrangement is in the first row b,

Gradient Descent

- start with some w, b . Keep changing w, b to reduce $J(w, b)$ until we settle at or a near minimum.
- ARTIFACT: image of one going to the point in a valley (gradient descent-surface plot). Another 2D plot ($b = 0$) to see the direction w is moving when changing the partial derivative and alpha. So they can see that if alpha is too small, they need a lot of baby steps and if it is too big, they might skip the local minima (fail to converge / diverge). also show that when it approaches to local minima, it reduces the size of the step automatically and when it reaches the local minima the partial derivative is 0 so it moves no longer.
- W is updated to the old value of w minus alpha times d over dw of the cost function.
- alpha and the term are the steps. Alpha is the learning rate - controls how big is the step. There a derivative term of the cost function (derivative of the cost function with respect to w).

Multiple Linear Regression

- Show the new equation with vectors and dot product (vectorization makes it computationally easy than the other way)
- vectorization = parallel computing

Font Typography

Arapey
NeuroLab
NEUROLab
NeuroLAB

Poppins Light

Aa

Aa Bb Cc Dd Ee Ff Gg Hh Ii Jj Kk Ll Mm Nn
Oo Pp Qq Rr Ss Tt Uu Vv Ww Xx Yy Zz

1, 2, 3, 4, 5, 6, 7, 8, 9, 0

Aa Bb Cc Dd Ee Ff Gg Hh Ii Jj Kk Ll Mm Nn Oo
Pp Qq Rr Ss Tt Uu Vv Ww Xx Yy Zz

1, 2, 3, 4, 5, 6, 7, 8, 9, 0

Color Palette

#154C9B

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